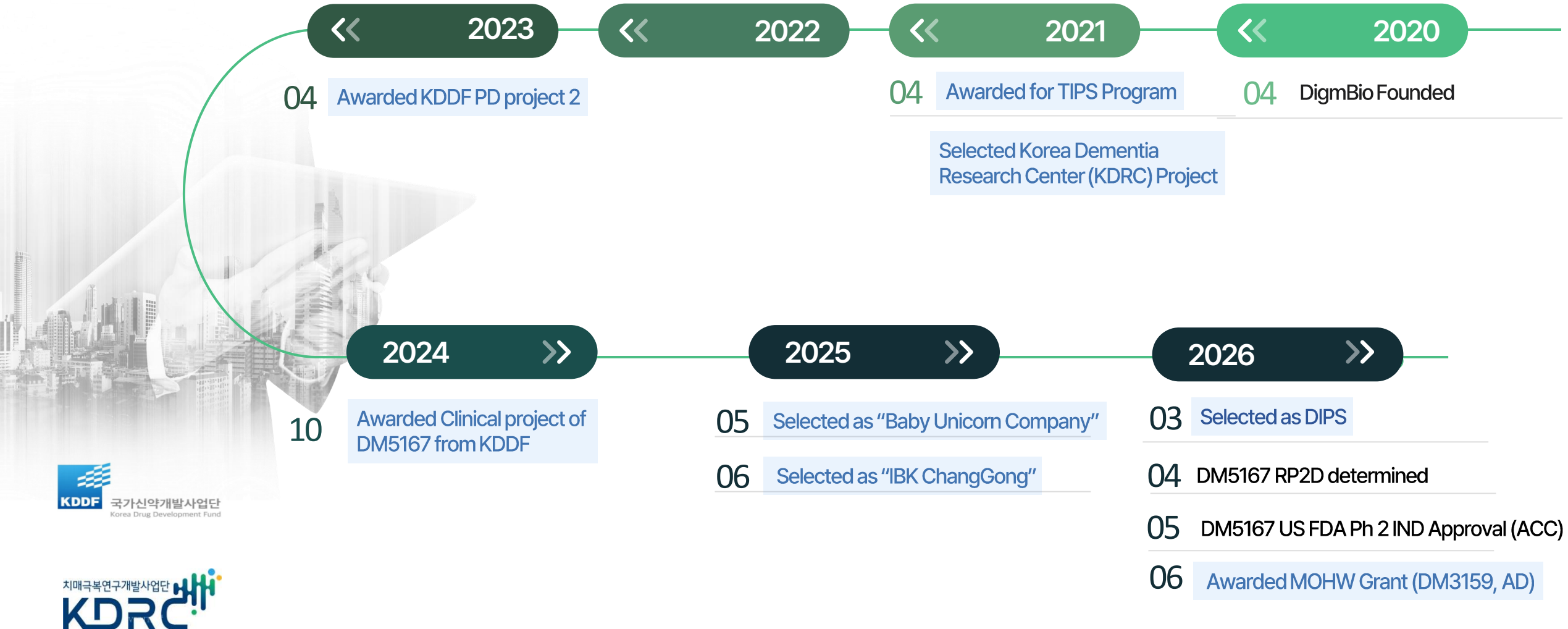


DigmBio

Company Name	Digmbio Inc
Established	2020. 04.
CEO	Jeongmin Kim
Employee	15 (11 R&D scientists / 8 Ph.Ds)
HQ	Seongnam-si, South Korea
Main Research Area	Neurodegenerative Disease and Cancr
Stake Holders	Jeongmin Kim & Related parties (46.1%), Investment firms (46.48%), Individual investors (7.42%)
Homepage	www.digmbio.com

Company History

| Beyond paraDigm-shifting company



Company Leadership



CEO
JeongMin Kim,
PhD

- Acclaimed for his prolific career in pioneering new drug R&D endeavors across esteemed Korean pharmaceutical entities like LG Chemical, Green Cross Biopharma, and Jeil Pharmaceutical Co.
- Ph.D. in biochemistry from the University of Wisconsin-Madison.



CSO
Jung-Ho Kim,
PhD

- One of the development members of Lazertinib, the 3rd generation EGFR inhibitor, while at Oscotec Inc. Lazertinib was approved by the FDA in combination with Amivantamab by J&J last year.
- Ph.D. in chemistry from the University of Illinois at Urbana-Champaign and completed his postdoctoral fellowship at Stanford University.



CCO
Young-Joo AHN,
Pharmacist

- Served as Head of Medical Division at JW pharmaceutical, Handok and Green Cross Biopharma.
- Master's Degree in Clinical Pharmacy from Sookmyung Women's University



CFO
Seong-Ryong Hwang

- 34 years of experiences in finance and investment from Daewoo Securities (Miraeasset securities)
- CFA charter obtained in 2002
- Master's Degree in English literature from Seoul National University.



Managing Director,
Bio Division
Myung-Eun Jung,
PhD

- Held exclusive leadership roles and spearheaded various R&D initiatives, spanning small molecule, vaccines, plasma derivatives, and antibody therapies at Green Cross Biopharma for about 15 years.
- Ph.D. in Neuroscience from Hallym University.



CBO
Gyu Hyun (George) Nam,
PhD

- 23 years of experiences in business development & marketing in local pharma (Hyundai Pharm), MNC (Otsuka), Samsung Bio & Biotech (TiumBio, AimedBio, etc.)
- IPO (KOSDAQ) in TiumBio.
- Ph.D. in Protein Engineering from POSTECH.

DM3159 (TASR Platform) – Alzheimer's Disease

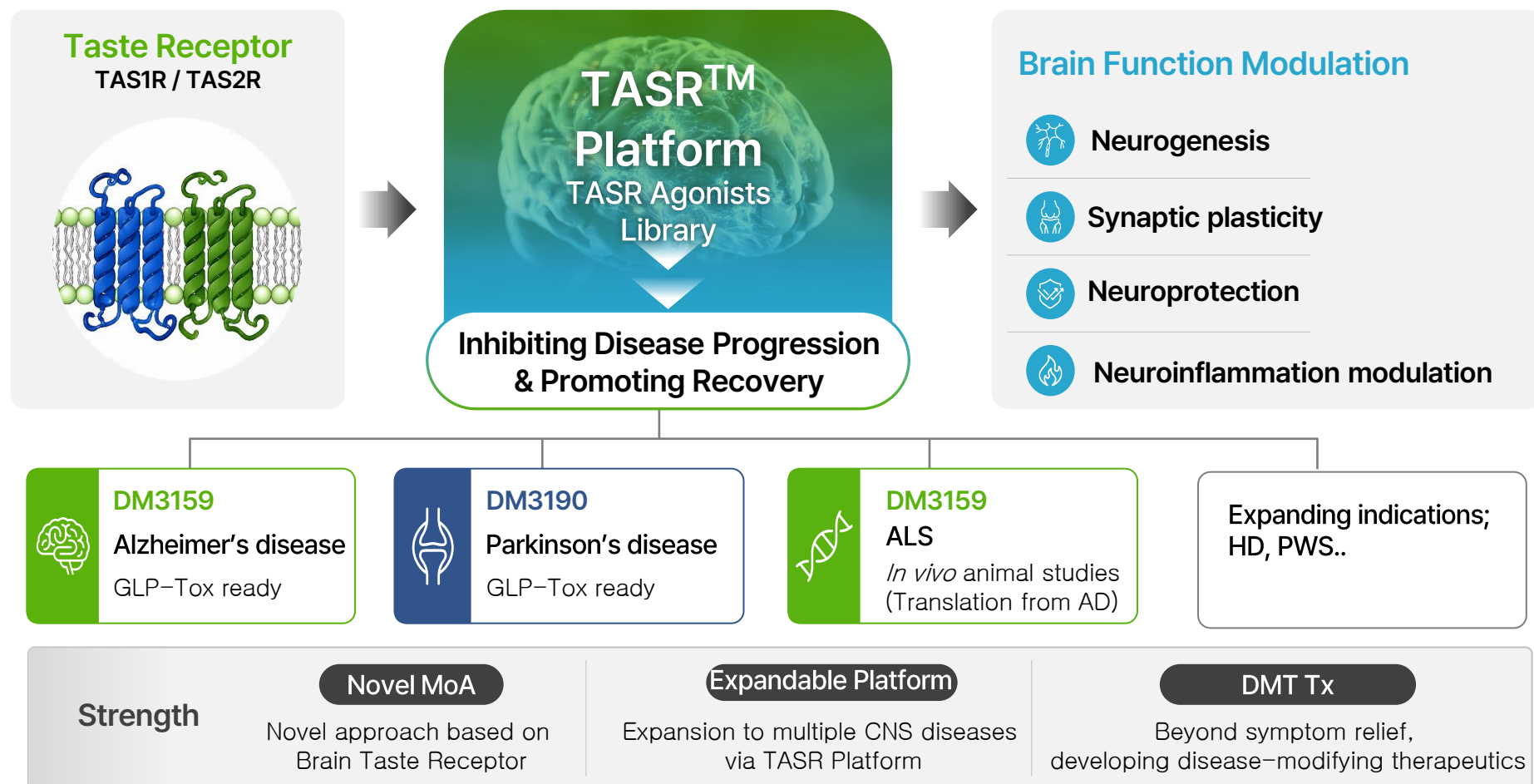


| Neurodegenerative Diseases (AD, PD, HD, ALS)

- ✓ **DM 3159** has demonstrated **outstanding efficacy** as a drug candidate for Alzheimer's disease in mouse animal models
- ✓ Very **high brain penetration rate** – 86% for DM3159 (based on rodents).

TASR (Taste Receptor) Platform

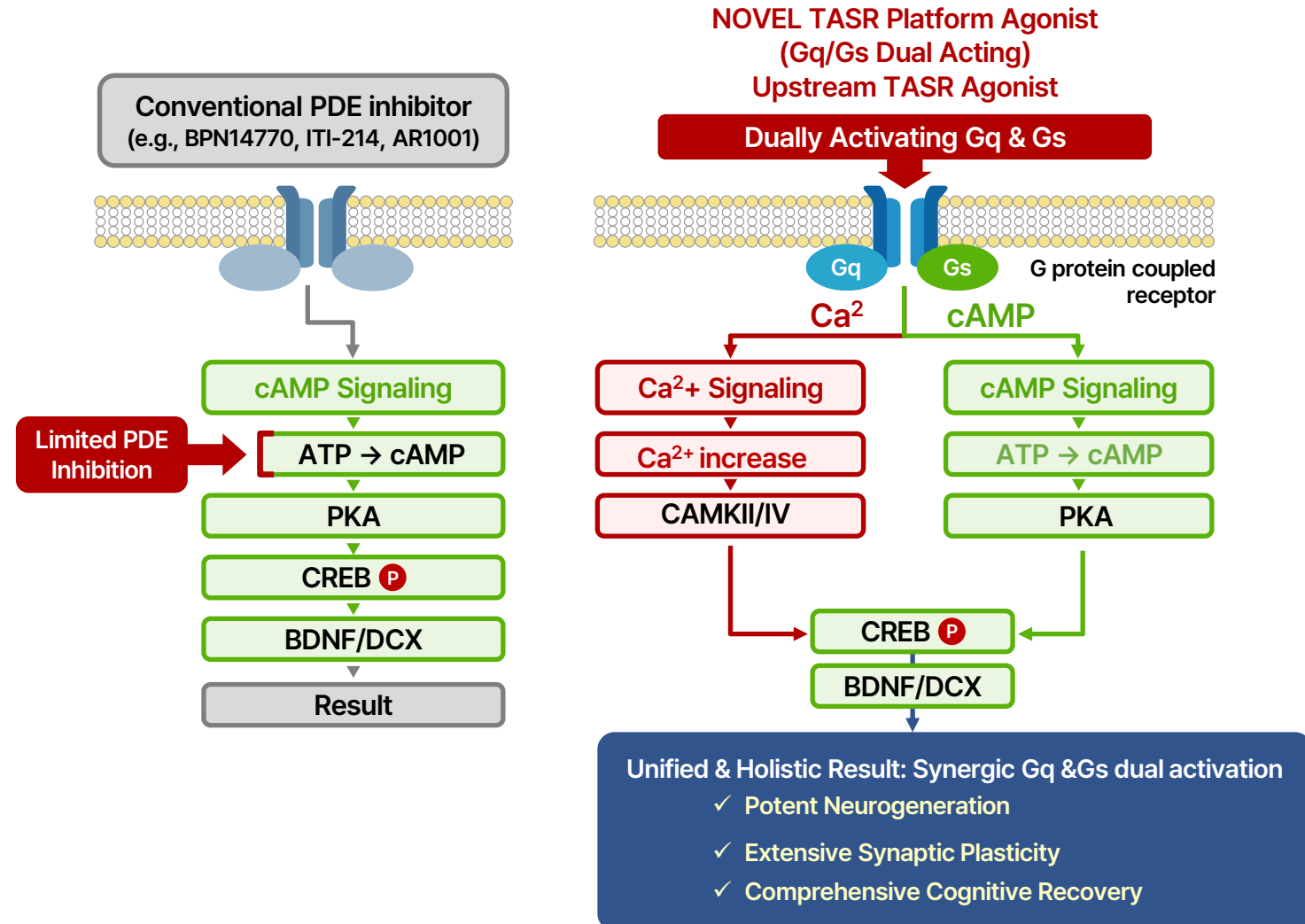
A Breakthrough CNS Drug Discovery Platform: Harnessing Brain Taste Receptor



TASR – MoA, Novel Dual Activation of Gq & Gs in GPCR

TASR Platform

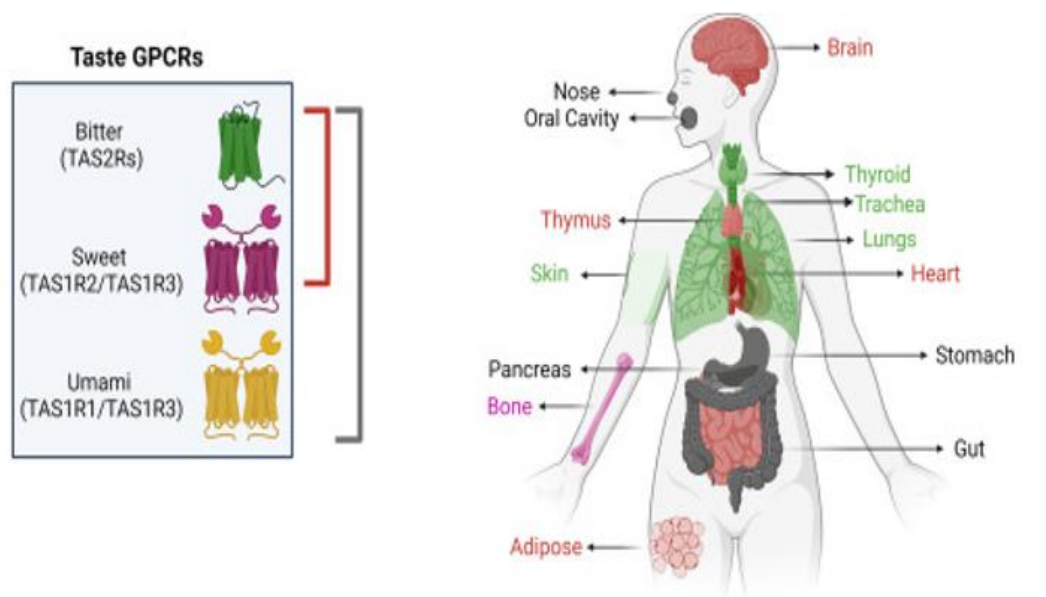
Mechanistic Analysis:
Novel **Dual**-Pathway TASR Agonist



T1R3 – Novel Target for Alzheimer's Disease

- Evolution of TASR research into brain functional receptors
- T1R as a promising therapeutic target for Alzheimer's disease

● Taste Receptors (TASRs)



● TASR – brain functional research

T1R3 & Alzheimer's disease

- Impaired perception of sweet and umami taste in AD patients
- Correlation between T1R3 function impairment and cognition impairment

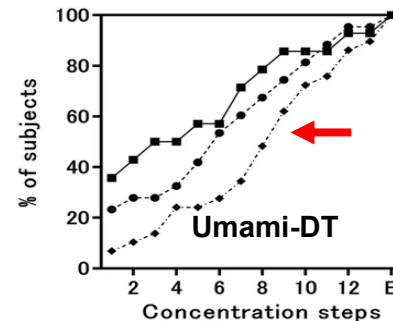
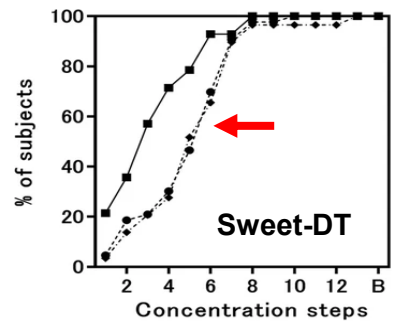
Organ/Tissue	RNA Expression (nTPM)
Brain	4.1
Salivary gland	7.7
Spleen	3.6
Small intestine	1.9
Duodenum	1.4
Lymph node	1.3
Esophagus	1.3
Heart muscle	1.2
Colon	1.0
Stomach	0.7
Kidney	0.5
Lung	0.4
Adrenal gland	0.3

[Source: The **Human** Protein Atlas]

MoA: TASR Agonist for Alzheimer's Disease

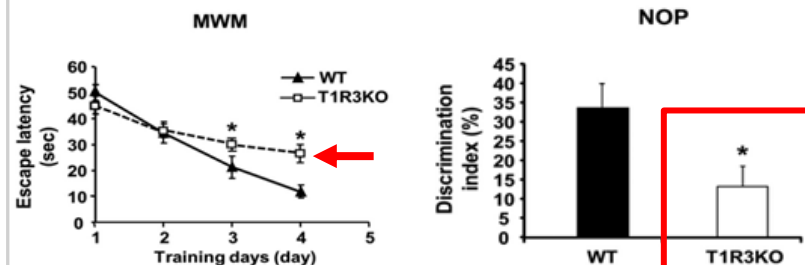
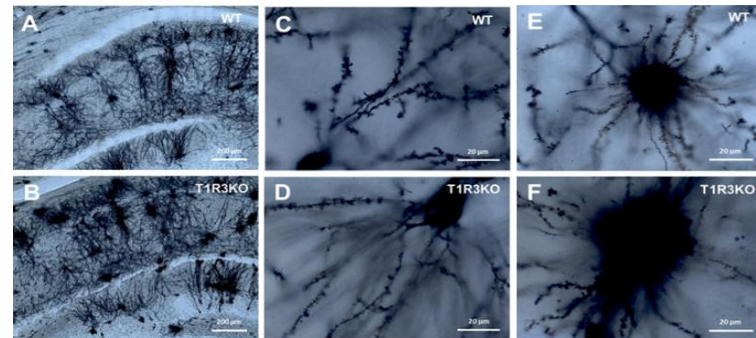
- Epidemiologically AD patients have a **low perception of sweet and umami tastes**, unlike normal people which means there is a close relation between T1R3 and AD.
- T1R3 KO mouse shows similar **hippocampal damage and learning & memory deficiency** to AD. T1R3 agonist dosing improves learning and memory and leads to an increase in synapsis and neurogenesis markers.

- AD patients have low perception of sweet and umami tastes which are coming from lowered brain T1R3 function and expression



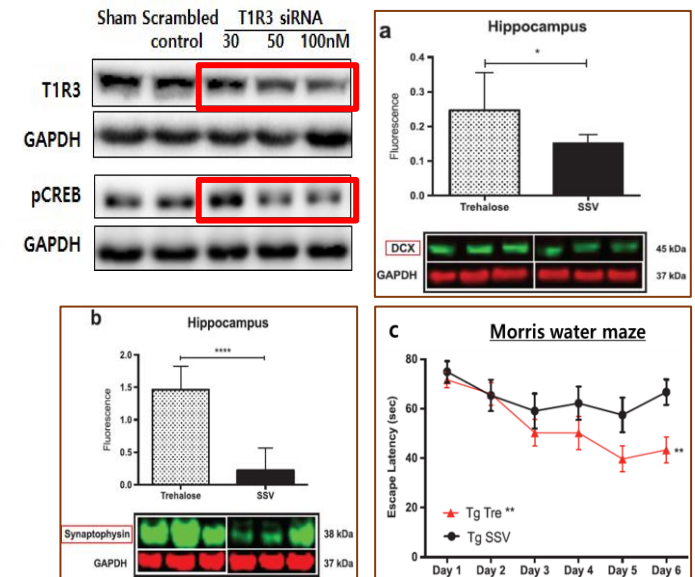
Reference: *BMC Neurol* 2020;20:110

- T1R3 KO mouse is different from normal mouse and shows damage in the hippocampus and learning memory deficiency similar to AD animal model



Reference: *J. Biol. Chem.* 2017 292(27) 11508–11530

- Animal with lowly expressed T1R3 shows decreases in p-CREB, synaptophysin, and DCX in the hippocampus which are related to learning and memory



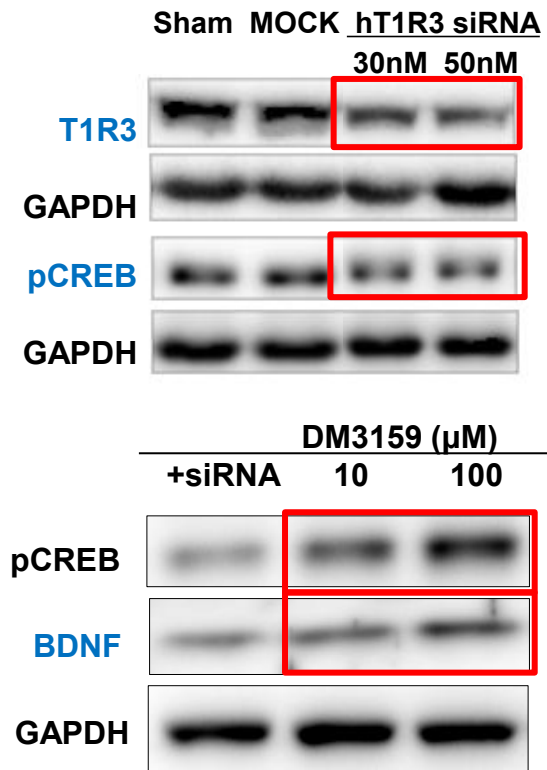
Synaptophysin(synaptic vesicle marker)

DCX(doublecortin,Marker of neurogenesis in the hippocampus)

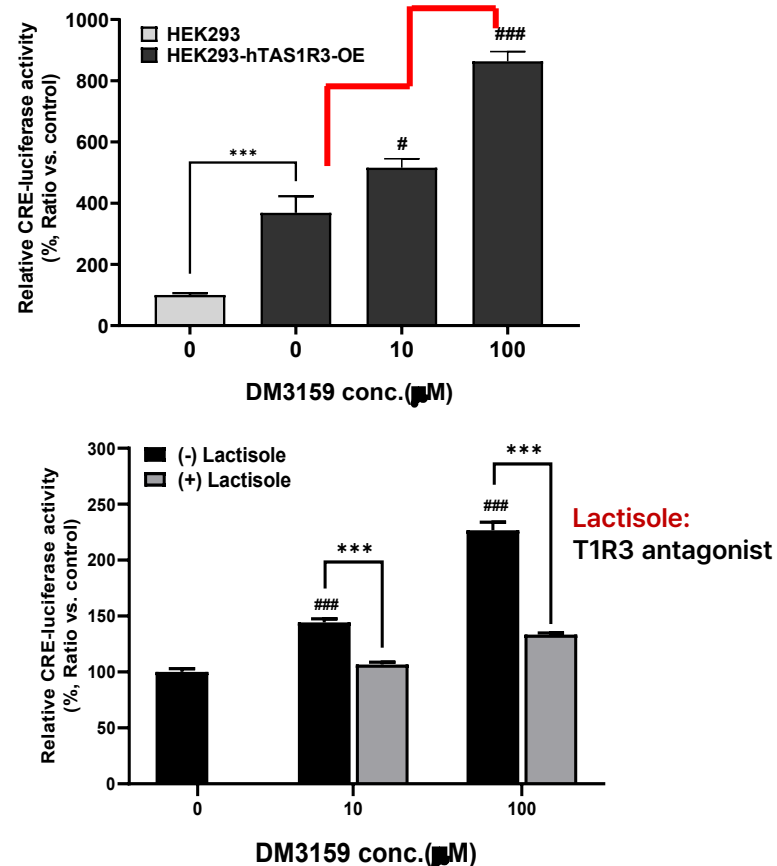
DM3159: pCREB/BDNF increase in T1R3

- DM3159 treatment upregulated **pCREB & BDNF** in T1R3-overexpression (OE) cell
- Confirming its potential for T1R3-mediated signaling activation

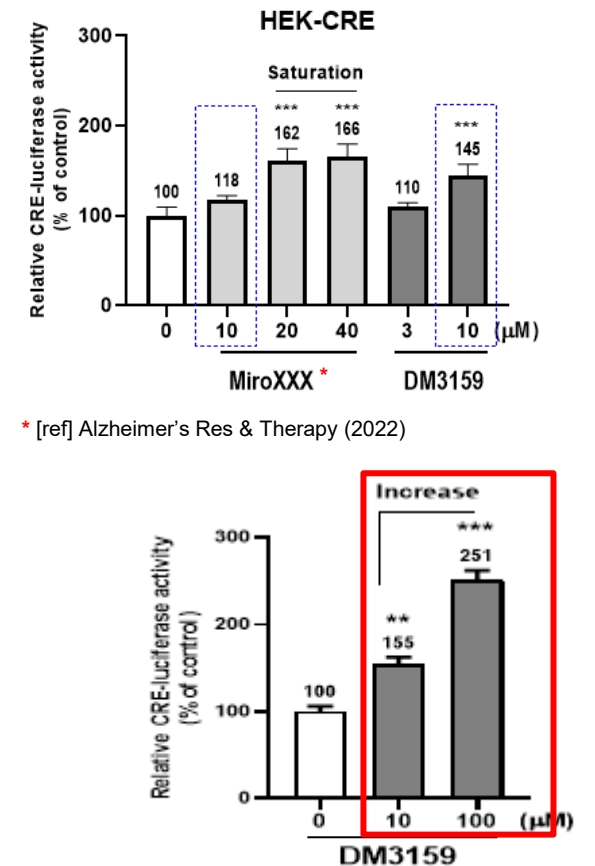
Effect of DM3159 in TAS1R3 siRNA



Effect of DM3159 in TAS1R3 overexpression (OE) cell

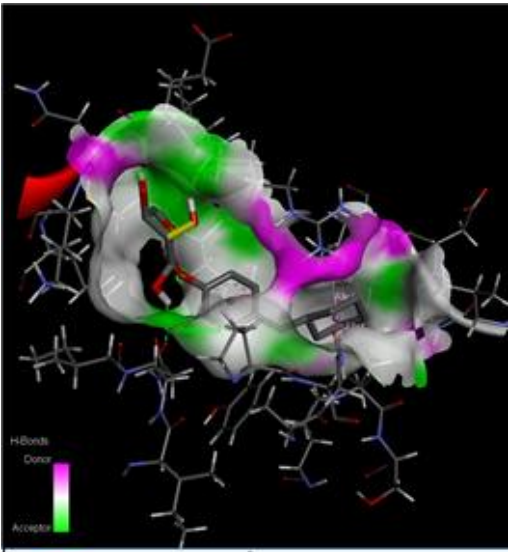


DM3159 vs PDE inhibitor

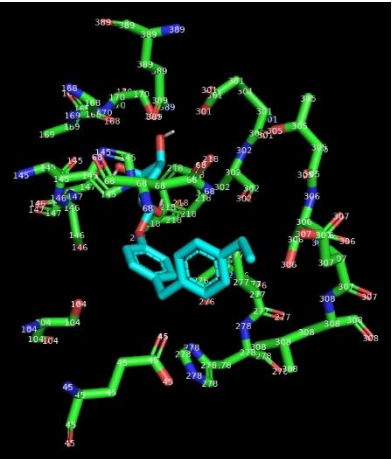


TASR Agonist: Discovery of Pre-clinical Candidates (~700 Library)

- Drug Design: AI & HIF1a surrogate marker *in vitro* screening



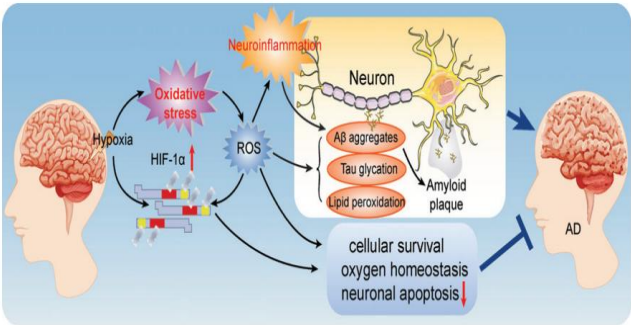
Binding pocket information of 3D TAS1R



Ki value	TASR
(uM)	0.108

Binding affinity of DM3159 for TASR

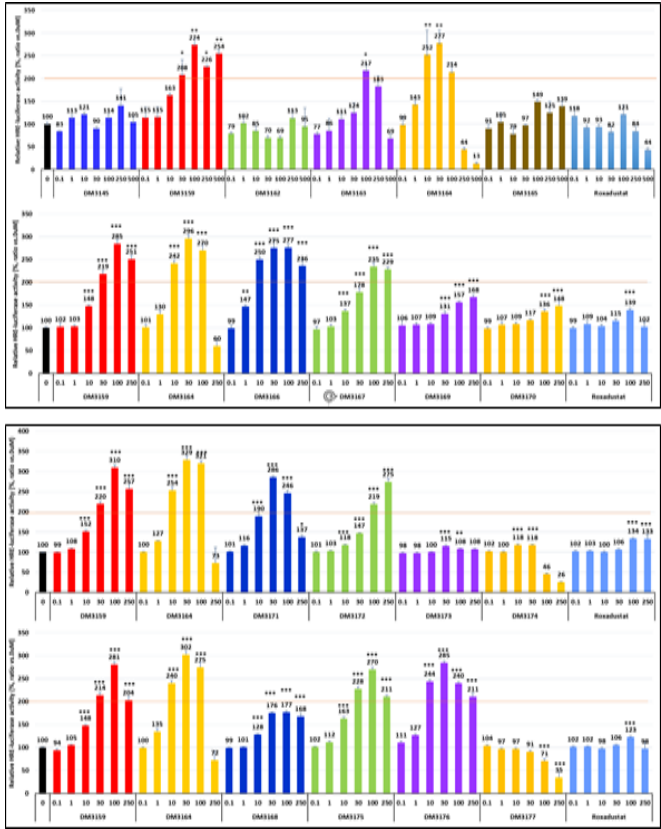
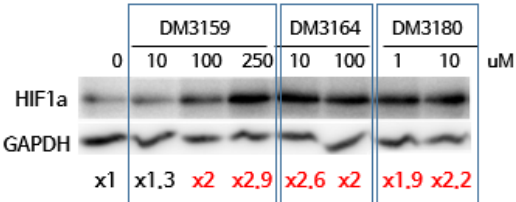
(a)



(b)

	HRE/luc (fold vs No treat.)
DM3159	2.7~3.2

Western blot



Target Product Profiling (DM3159)

	Contents		DM3159 (AD)
Efficacy	<i>In vitro</i>	Binding affinity ((kcal/mol)	T1R3(-8.3)
		HRE-luciferase activity	1.7~3-fold increase ($\geq 10\mu\text{M}$)
		Neuroprotection ($\text{A}\beta 1-42$)	$\geq 10\mu\text{M}$
		Neuroprotection (α -syn)	-
		Neuroinflammation (LPS)	$\geq 10\mu\text{M}$
	<i>In vivo</i>	Scopolamine-Induced AD Models	Excellent improvement (50mg/kg)
		3xTg & 5xFAD AD Models	Excellent cognitive improvement and Biomarker changes at 50mg
		Zebrafish PD Model	-
		α -Syn/PFF-induced PD Model	-
ADME/PK	CYP inhibition	CYP inhibition (IC_{50})	CYP1A2, 2B6, 2D6, 2C9, 3A4: $>50\mu\text{M}$
	Metabolic Stability	Liver microsomal stability	Stable ($T_{1/2}>60\text{min}$)
	Pharmacokinetics (PK)	AUC_{last} ($\mu\text{g}\cdot\text{hr}/\text{kg}$)/BA (%)	(Rat) $\text{AUC}=65.82$, BA=88.8% (Dog) $\text{AUC}=18.53$, BA=42.8%
	BBB Permeability	Brain-to-Plasma (B/P) Ratio	B/P = 0.86
Safety	Genotoxicity	Mini-Ames Test	Negative
	Cardiovascular safety	hERG/Dog telemetry	No effect
	General Toxicology	2weeks Repeat-Dose toxicity	(Rat) No significant toxicity at 200 mg/kg
	Off target toxicity	Safety screen44 panel assay	No significant inhibition

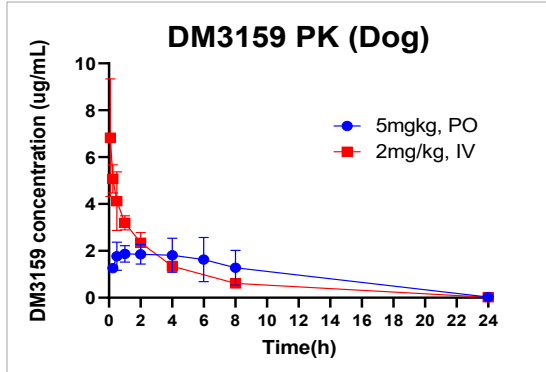
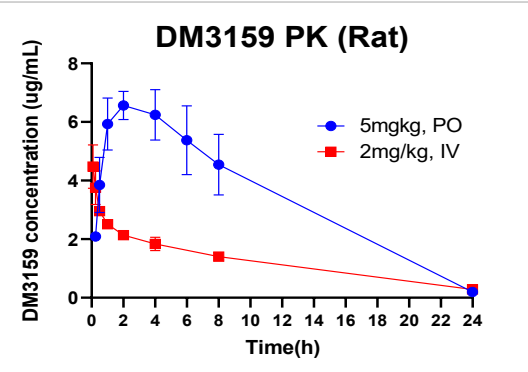
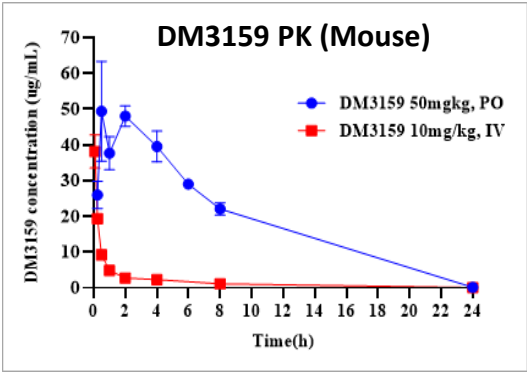
DM3159 (AD): *in vivo* PK and BBB Profile

- ✓ PK parameter analyzed in ICR mouse (male, PO)
- ✓ DM3159 showed **high BA** and **robust BBB permeability**

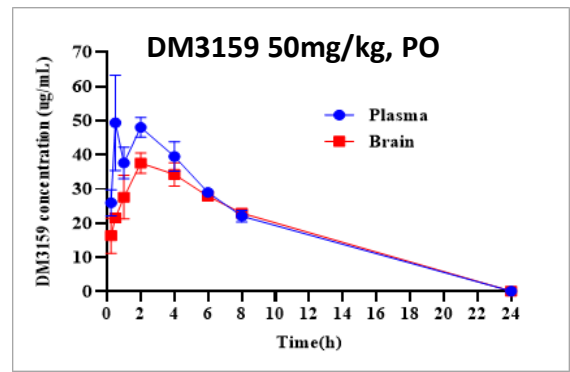
In vivo PK	Unit	Mouse		Rat		Dog	
		IV	PO	IV	PO	IV	PO
Dose	mg/kg	10	50	2	5	2	5
Cmax	ug/mL	38.1	49.4	4.89	6.78	8.05	2.27
Tmax	h	0.08	0.5	0.08	2.33	0.08	2.5
t1/2	h	3.9	2.4	7.36	3.60	3.23	6.01
AUClast	h*ug/mL	36.7	355.0	27.16	65.82	17.35	18.53
BA	%	192		88.8		42.8	

In vivo BBB	Unit	DM3159 (PO)	
		Brain	Plasma
Dose	mg/kg	50	50
Cmax	ug/mL	37.6	49.4
Tmax	h	2.0	0.5
t1/2	h	2.3	2.4
AUClast	h*ug/mL	305.4	355.0
AUC brain/plasma		0.86	

In vivo PK



In vivo mouse BBB



DM3159 (AD): Cognitive Improvement in 3xTG-AD Animal Model

7-month-old 3xTG mice (APP, PS1, Tau) were treated with DM3159 orally for 20 weeks, followed by Morris water maze test

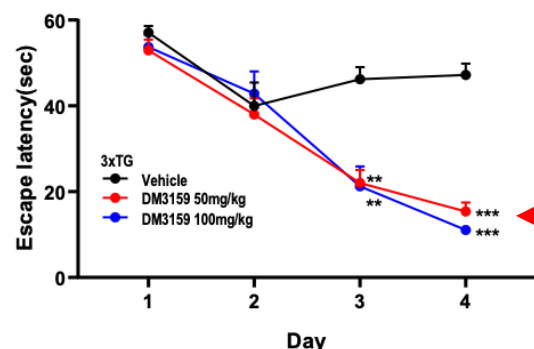
- DM3159 significantly improved cognitive impairment in 3xTG AD mice and enhanced cognitive function even in aged normal mice(11 months old)

Morris water-maze test

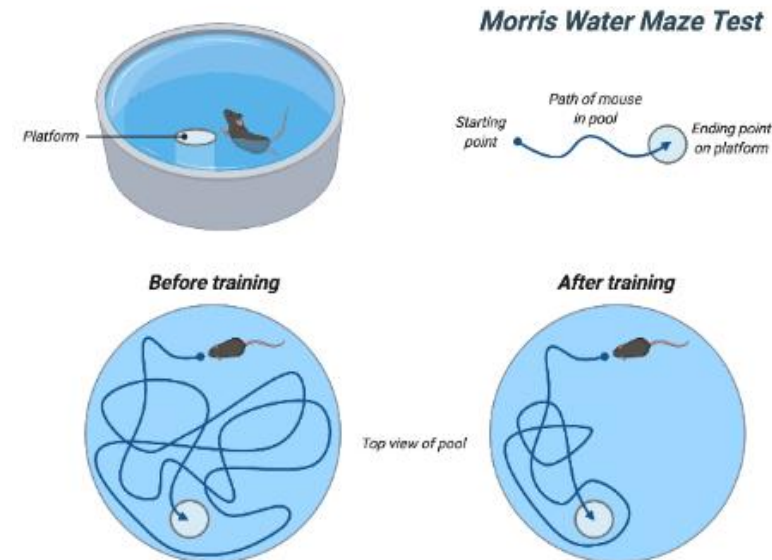
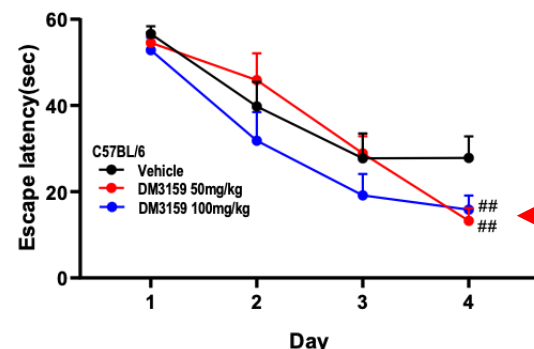
escape latency



11-month 3xTG AD mouse



11-month normal mouse

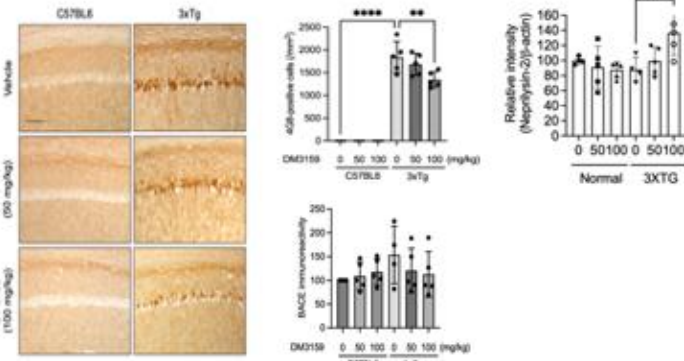


P values for the group comparisons were obtained by one way ANOVA followed by Student-Newman-Keuls test. ### p<0.001, # p<0.05 compared with the C57BL/6 Vehicle group. ** p<0.01, ***p<0.001 compared with the 3xTG Vehicle group.

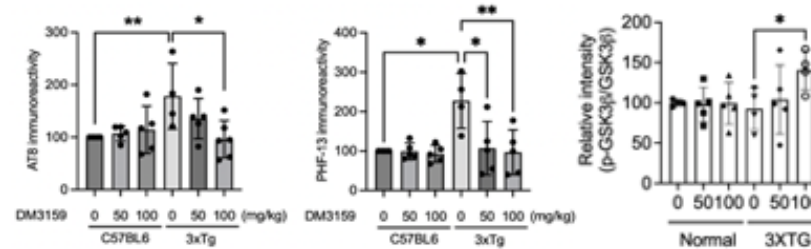
DM3159: Efficacious Changes in Key Brain Biomarkers in 20wk-treated 3xTG-AD Mouse

7-month-old 3xTG mice (APP, PS1, Tau) were treated with DM3159 orally for 20 weeks, followed by Morris water maze test and biomarker analysis

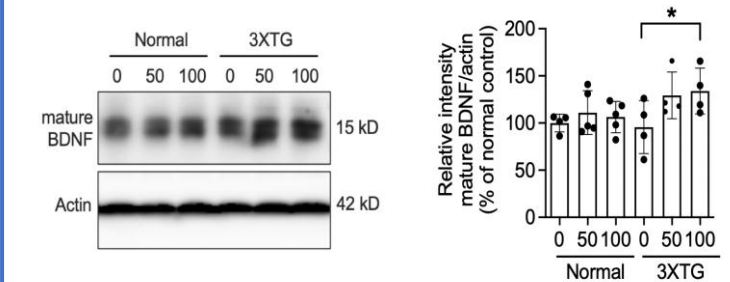
Reduced **A β** deposition, and **BACE** levels.
Increased Neprilysin-2-mediated **A β** clearance



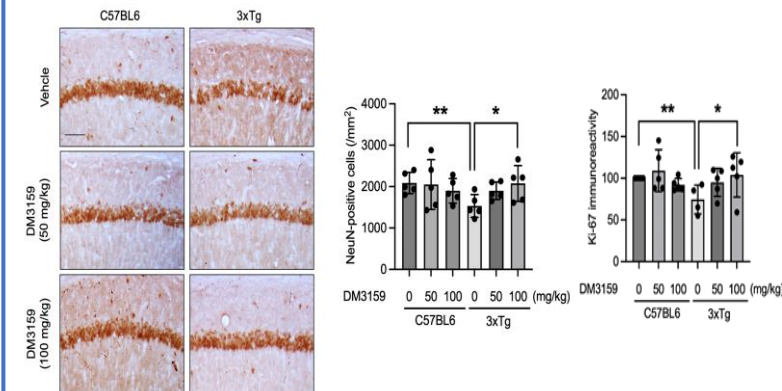
Reduced pathological **Tau** phosphorylation (AT8, PHF-13) Enhanced **GSK3 β** (Ser9) phosphorylation, inhibiting **Tau**.



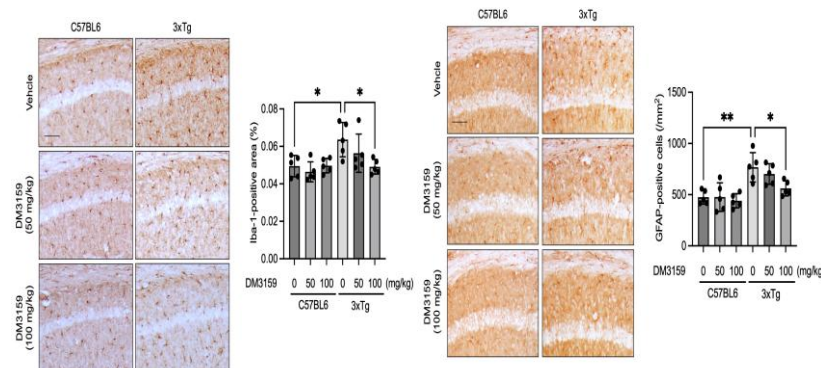
Increased **BDNF** level



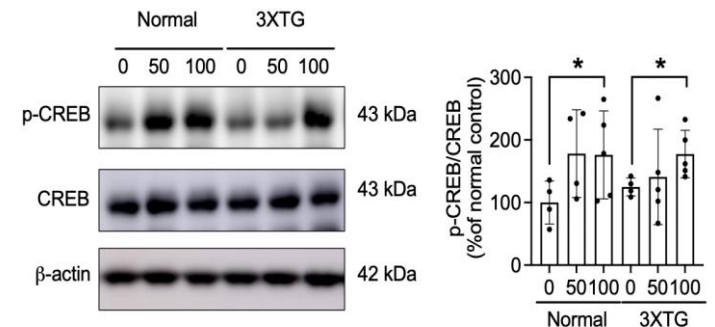
Enhanced neuronal generation (NeuN) and proliferation (Ki67)



Reduced microglial and astrocyte activation



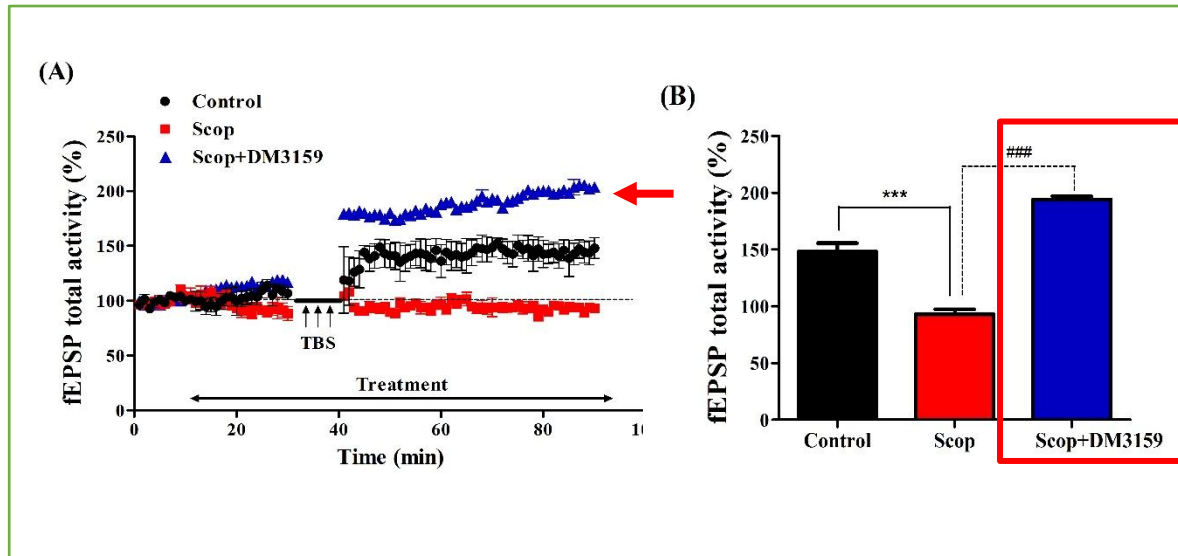
Increased **Synaptic plasticity (pCREB)** level



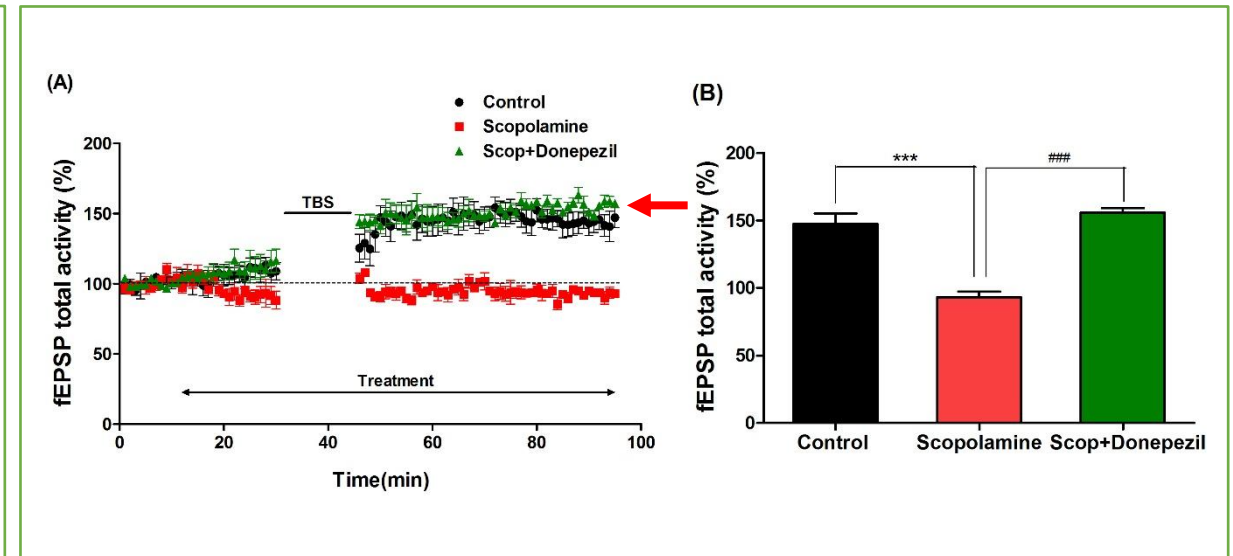
DM3159: Reversing Scopolamine-induced LTP Impairment

- Scopolamine-induced LTP (Long-Term Potential) impairment is a model of synaptic dysfunction rather than Alzheimer's disease itself.
- DM3159 fully **restored impaired LTP**, demonstrating **robust recovery of synaptic function**. A reference experiment with donepezil also restored LTP to the normal range.

■ DM3159 experiment



■ Reference. Donepezil experiment



Independent experiments performed using the same hippocampal slice LTP protocol.
Data are shown for qualitative reference only.

DM3159 (AD): Safety Pharmacology

- ✓ No drug-related cardiovascular effects across the tested dose range by telemetry-based CV parameter evaluation
- ✓ No significant effect on hERG & off-target toxicity, Ames test (-), good tolerability in 2-week repeated dose tox study

Parameters	DM3159 (PO)	
Dose	50 mg/kg	100 mg/kg
Heart rate, Systolic blood pressure, Diastolic blood pressure, Mean blood pressure, Body temperature	Normal	Normal
PR interval, QRS duration QT interval, QTc interval	Normal	Normal

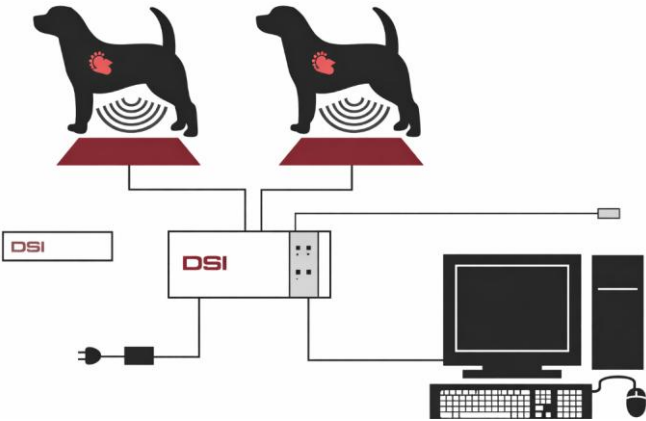


Fig. Safety Pharmacology Test in Beagle dog

Parameters	DM3159 (PO)
CYP inhibition (IC ₅₀)	CYP1A2, 2B6, 2D6, 2C9, 3A4 >50uM
hERG assay (10uM)	No significant inhibition
Mini-Ames (TA98, TA100 strain)	Negative
Repeated does Toxicity	SD Rat, 200 mg/kg, PO, 2-week repeated dose (non-GLP)
	No clinical signs observed. No abnormalities in hematology or clinical chemistry. Mild splenic extramedullary hematopoiesis observed in histopathology.
Off target Tocixity (Safety screen44 panel assay (10uM)	No significant inhibition

DM3159(AD): Off-Target Effect

- ✓ No significant inhibition (>50% inhibition) in Safety screen44 panel assay
- ✓ DM3159 showed **low off-target effect**

	Safety screen44 panel	%-Inhi. (in 10uM)
		DM3159
1	Cholinesterase, Acetyl, ACES	2
2	Cyclooxygenase COX-1	0
3	Cyclooxygenase COX-2	1
4	Monoamine Oxidase MAO-A	16
5	Phosphodiesterase PDE3A	0
6	Phosphodiesterase PDE4D2	3
7	Protein Tyrosine Kinase, LCK	26
8	Adenosine A2A	3
9	Adrenergic α 1A	-1
10	Adrenergic α 2A	3
11	Adrenergic β 1	-7
12	Adrenergic β 2	-2
13	Androgen (Testosterone)	-13
14	Calcium Channel L-Type, Dihydropyridine	3
15	Cannabinoid CB1	4

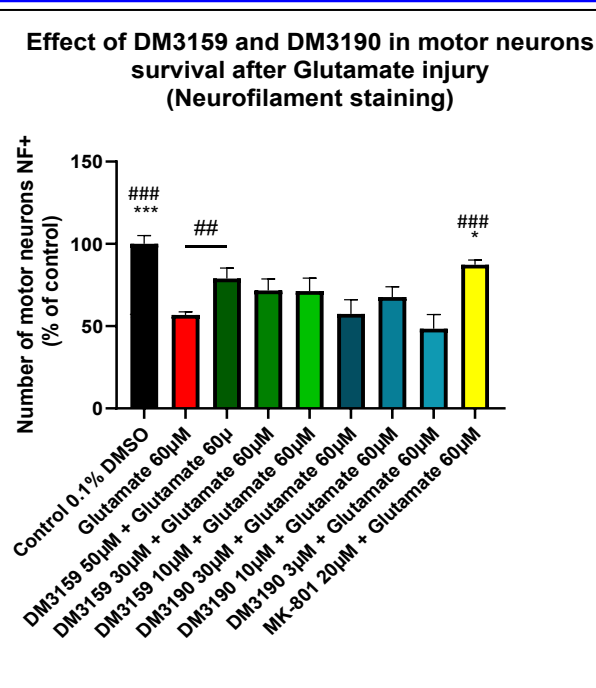
	Safety screen44 panel	%-Inhi. (in 10uM)
		DM3159
16	Cannabinoid CB2	-10
17	Cholecystokinin CCK1 (CCKA)	-12
18	Dopamine D1	3
19	Dopamine D2S	4
20	Endothelin ETA	-6
21	GABAA, Flunitrazepam, Central	-1
22	Glucocorticoid	9
23	Glutamate, NMDA, Agonism	3
24	Histamine H1	12
25	Histamine H2	-13
26	Muscarinic M1	3
27	Muscarinic M2	6
28	Muscarinic M3	6
29	Nicotinic Acetylcholine α 4 β 2, Cytisine	7
30	Opiate δ 1 (OP1, DOP)	-3

	Safety screen44 panel	%-Inhi. (in 10uM)
		DM3159
31	Opiate κ (OP2, KOP)	5
32	Opiate μ (OP3, MOP)	-8
33	Potassium Channel [KA]	8
34	Potassium Channel hERG, [3H]Dofetilide	11
35	Serotonin 5-HT1A	5
36	Serotonin 5-HT1B	-2
37	Serotonin 5-HT2A	11
38	Serotonin 5-HT2B	12
39	Serotonin 5-HT3	5
40	Sodium Channel, Site 2	-5
41	Transporter, Dopamine (DAT)	12
42	Transporter, Norepinephrine (NET)	12
43	Transporter, Serotonin (SERT)	4
44	Vasopressin V1A	-4

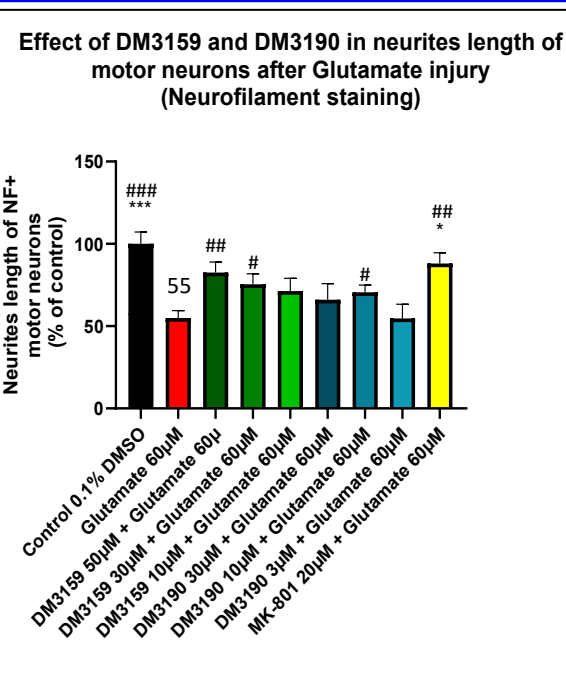
ALS (Amyotrophic lateral Sclerosis), *In Vitro* Data

- DM3159 significantly **protects motor neurons** from glutamate-induced neurotoxicity and preserves neurite integrity in a primary ALS cell model.

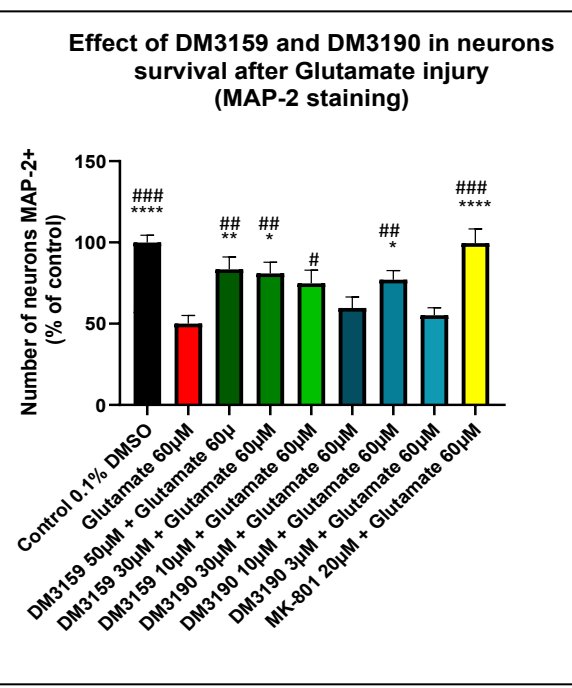
DM3159 improves neuronal survival



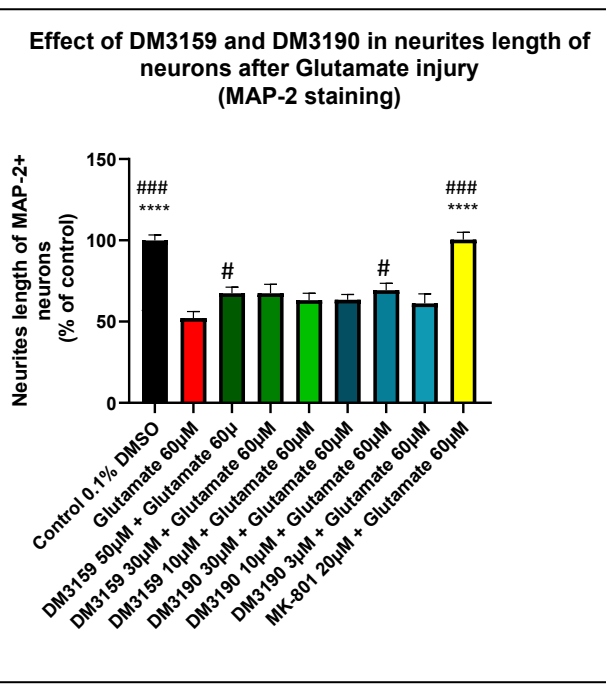
DM3159 preserves neurite length



DM3159 improves neuronal survival



DM3159 maintains neurite integrity



CRO: Neuron expert

- * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$ (vs Glutamate), one-way ANOVA
- # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$ (vs Glutamate), student's t-test
- NF+ (neurofilament) → mature motor neurons, axon staining
- MAP2 (Microtubule associated protein 2) → motor neurons, interneurons and immature neurons, dendrite staining

DM3190 (PD) is effective for Parkinson's Disease

α -Synuclein/PFF induced mouse PD model

- DM3190 showed **significant behavioral improvements** comparable to **[L-DOPA plus Rasagiline]** in an α -synuclein/PFF-induced mouse model of Parkinson's disease.

Cylinder test

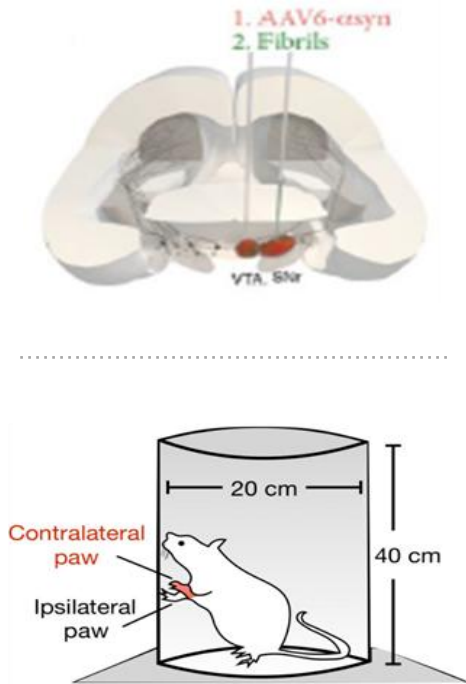
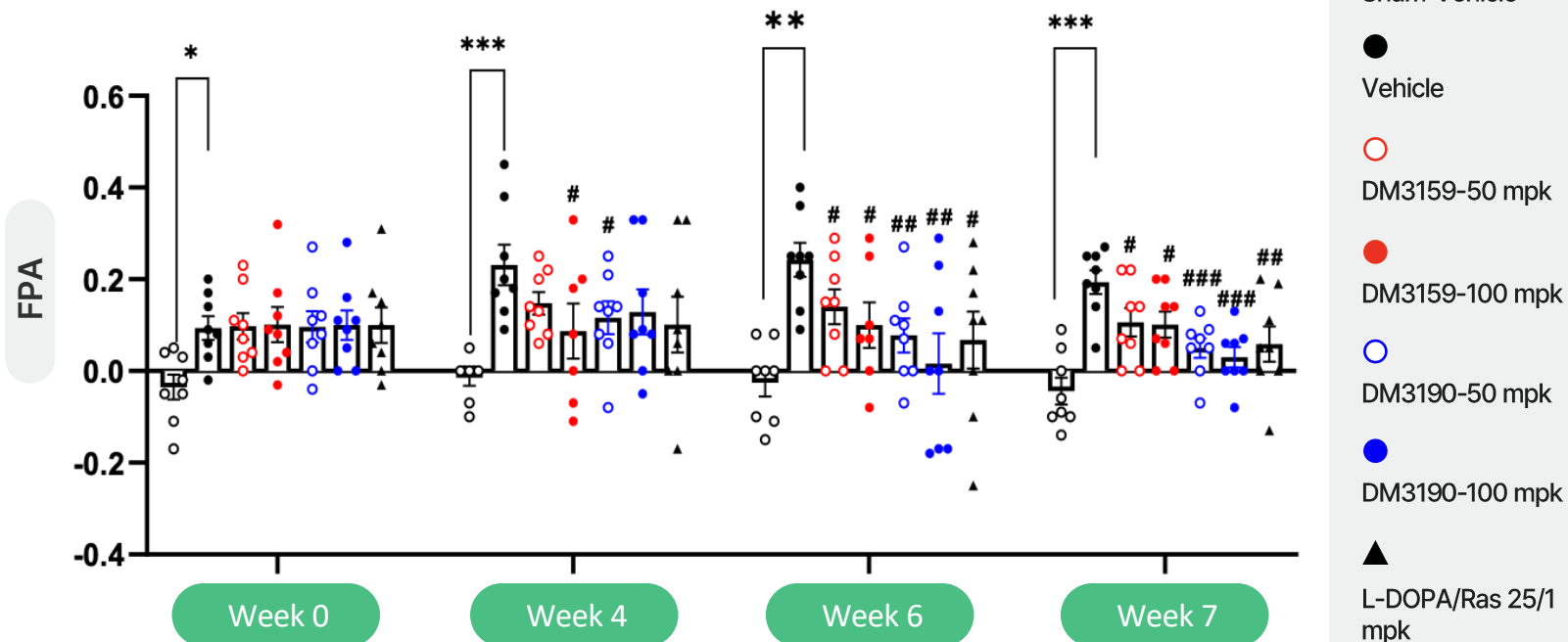
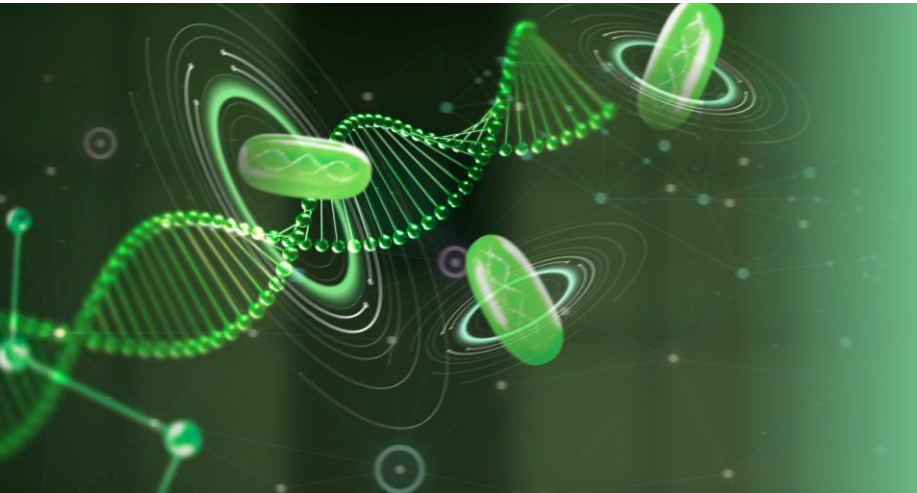


Fig. Cylinder test

[R&D Pipelines – CNS] Development Timeline

Disease Area	Code Number	Modality	Indications	Timeline (Year)					
				2026	2027	2028	2029	2030	
Oncology	DM5167	Chemical	Solid Tumors	P1		P2			
			Brain Tumors	P1		P2			
			ACC		P2				
	DM7105	PROTAC	Solid Tumors	Candidate	Pre-Clinic		P1		
CNS	DM3159	Chemical	AD	Pre-Clinic	P1		P2		
	DM3159	Chemical	ALS (HD)	Pre-Clinic		P1			
	DM3190	Chemical	PD	Pre-Clinic	P1		P2		

DM3159 (TASR Agonist): Executive Summary



- ✓ TASR Agonists, DM3159: Highly Potent Neurogenic Agents
- ✓ **DM 3159** demonstrated world-leading cognitive enhancement in 3xTg-AD mouse models
 - Pathology normalization: effectively target b-amyloid, phospho-tau, and microglial activity
 - Synaptic plasticity: promotes symptom relief by elevating pCREB and BDNF levels
- ✓ TASR platform enabled the simultaneous discovery of therapeutic candidates to expand Parkinson's disease, ALS, HD, FXS
 - Extensible to orphan diseases, including ALS (Lou Gehrig's), Huntington's disease, and Fragile X Syndrome

Thank you for your attention!



Contact:

Gyu Hyun (George) Nam, Ph.D., SVP/CBO

Digmbio Inc.

901, 9-22 Pangyo-ro 255beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do, Korea

E-mail) george@digmbio.com

Phone) 82-10-8513-8820